

Linguistic Echoes: Explainable NLP for Academic Stress Detection

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QUADRANT 1

Research Question

Can explainable NLP recover meaningful psycholinguistic stress markers — and do different academic communities express distress in distinct ways?

- Do XAI methods recover known psycholinguistic stress markers independently?
- Which features best separate stressed from non-stressed writing?
- Do undergrad, grad, and pre-med students show distinct stress profiles?
- What are the ethical risks of automated stress detection in education?

Why it matters: 40%+ of university students report overwhelming anxiety. Most support is reactive. Explainable NLP can surface early warning signals and reveal how communities differ in expressing distress.

QUADRANT 2

Methodology / Project Design

- 1 **Data:** Dreddit (3,553 stress-annotated posts, balanced labels, 93 LIWC features) for training. 8,000 posts from r/college, r/GradSchool, r/PhD, r/PreMed (Arctic Shift) for inference.

2 **Two models trained on Dreddit:**

TF-IDF + Logistic Regression
Unigrams/bigrams · balanced weights
· interpretable coefficients

Fine-tuned DistilBERT
3 epochs · AdamW · Tesla T4 GPU (Kaggle)

3 **Explainability (XAI):**

LIWC psycholinguistic features

SHAP global attribution

LIME local explanation

4 **Statistical analysis:** Mann-Whitney U tests · Benjamini-Hochberg FDR correction (15 tests) · Cohen's d effect sizes · Spearman correlations
- QUADRANT 3
- Data Analysis & Results
- KEY DISCOVERY**
Undergraduate communities show significantly greater self-focused language; PhD communities show more relational language — all findings held after Benjamini-Hochberg correction.
- d=0.33**
College vs PreMed
1st-person effect size

0.874
DistilBERT ROC-AUC

p=0.53
1st-Person ↔ Stress
Prob

12/15
Significant after BH
correction
- XAI Convergence**
SHAP, LIME, and LIWC independently recovered the same markers: *me/my/myself* and *anxiety/scared/feel* predict stress; *you/we/her* predict non-stress.

Community Differences
r/college shows highest 1st-person singular rate (d=0.328 vs PreMed, p<0.001, BH-corrected). PhD students show more social words (d=-0.126), suggesting relational framing of distress.
- | Marker | TF-IDF | SHAP | LIME | LIWC |
|-------------------------|---|------------|------------|-----------|
| me / my / myself | ✓ Stress | ✓ Stress | ✓ Stress | lex_i ↑ |
| anxiety / scared / feel | ✓ Stress | ✓ Stress | ✓ Stress | lex_anx ↑ |
| you / we / her | Not stress | Not stress | Not stress | posemo ↑ |
| Anxiety vocab alone | Does NOT differentiate GradSchool vs PhD (p=0.677, ns after BH) | | | |
- QUADRANT 4
- Interpretation & Conclusions
- XAI recovers theory-grounded markers:** SHAP, LIME, and LIWC independently converged on first-person singular + negative emotion, matching Social Monitoring Theory.

Communities express stress differently: College shows highest self-focus (p<0.001, d=0.38). PhD students use more relational language. Anxiety vocab alone is not diagnostic.

19.3% false negative rate: Stress in third-person or retrospective voice is systematically missed.

Limitations: General Reddit data (domain shift) · Small-to-moderate effect sizes · No clinical ground-truth labels
- Ground-truth labeled academic writing
 - Temporal stress modeling across a semester
 - Fairness audit across demographic groups
 - Investigate PhD relational framing vs suppression